

Part B Report Outline

Caldwell University Student Dropout Project

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1. Executive Summary

Caldwell University faces a 25.6% withdrawal rate (115 of 450 students), creating both mission risk and roughly \$15,000 in lost tuition per departing student. The Dean of Student Success must decide which students should receive limited advising and outreach capacity first. This report proposes a two-stage analytical workflow: a Random Forest classifier screens for elevated dropout risk ($AUC = 0.744$), and a regression model estimates likely withdrawal timing among dropouts ($R^2 = 0.575$). The key finding is that dropout risk is not one problem: early dropouts show weaker academic and engagement signals, while late dropouts show stronger academic performance but greater work and scheduling pressure. We therefore recommend a tiered pilot that routes urgent academic cases to high-touch advising and later-stage risk to financial or scheduling support.

2. Business Problem and Decision Context

Caldwell's current early-warning process relies on mid-semester grade flags and advisor referrals, so it often acts late and treats flagged students too similarly. The operational decision is not simply whether a student is risky, but how to prioritize limited support by both intensity and timing. The cost structure is asymmetric: missing a true at-risk student can cost about \$15,000, while a false flag costs about \$500. Therefore, the model is designed as a recall-oriented screening tool, not an automated decision-maker.

In CoNVO terms, the context is constrained Student Success capacity; the need is earlier prioritization and urgency-based routing; the vision is a re-scorable term-by-term workflow; and the outcome is lower dropout in the flagged cohort with positive expected value. The tool must

remain interpretable, auditable, and subject to advisor review, especially because student-risk models can easily shift from support tools into unfair labeling systems if they are not governed carefully.

3. Dataset and Data Understanding

The dataset contains 450 current-year student records, one row per student. The outcome is dropout status: 115 students withdrew (25.6%) and 335 were retained. The predictors fall into three business-relevant signal groups. Academic variables, such as GPA, credits completed, and course withdrawals, show current performance. Behavioral variables, such as attendance and advisor meetings, indicate engagement. Situational variables, including enrollment type, financial aid, first-generation status, and employment hours, capture external pressure. This grouping supports the two-stage design: the classifier uses academic and behavioral signals to screen risk, while the timing regression helps distinguish urgent academic-risk cases from later-stage financial or scheduling-risk cases.

Table 1 shows the baseline pattern: dropout students have lower GPA, lower attendance, fewer completed credits, more course withdrawals, and higher work hours than retained students. The sharper insight, however, is that dropout students are not homogeneous. Splitting the dropout group at the median withdrawal point (49 credits) produces two profiles. Early dropouts had an average GPA of 2.02, 73% attendance, and 18.7 work hours per week, suggesting the need for tutoring, course-load review, and probation support. Late dropouts had a stronger average GPA of 2.47 and 79% attendance but worked 25.9 hours per week, suggesting that financial aid, scheduling flexibility, and part-time pathway support may be more useful. Part-time students also left 14 credits later on average than full-time students despite dropping out at a higher rate (38.9% vs. 20.4%), reinforcing that timing and intervention type should be separated.

Table 1

Mean Differences Between Retained and Dropout Students Across Academic, Behavioral, and Situational Variables

Variable	Retained (n = 335)	Dropout (n = 115)
Current GPA	2.93	2.24
Attendance Rate	0.82	0.76
Credits Completed	11.0	7.5
Course Withdrawals	0.79	1.51
Employment Hours/Week	19.5	22.3

Note. Values are sample means. N = 450.

Several data limitations shape interpretation. First-generation students drop out at 35.7% versus 21.6%, so using that variable requires fairness monitoring rather than automatic targeting. The dataset is structurally clean, with no missing values or outliers, but substantively incomplete because it excludes mental health, housing burden, family obligations, prior academic history, and advisor notes. These omissions create omitted-variable risk and limit causal interpretation. Categorical variables were one-hot encoded; the 80/20 stratified split preserves the dropout rate across training and test sets. Overall, the dataset is useful for screening and triage, but advisor judgment remains necessary before intervention decisions.

4. Analytical Framework and Methodology

This analysis uses a two-stage design to address the Dean of Student Success's resource allocation decision: Stage 1 identifies which students are at elevated dropout risk (classification); Stage 2 estimates how far along flagged students are likely to be when they withdraw (regression), enabling urgency-based prioritization.

4.1 Stage 1: Classification Model

Random Forest was selected over Logistic Regression because it captures non-linear interactions, handles class imbalance through balanced weighting, and produces feature importance scores that support advising decisions. Recall is the primary metric — missing an at-risk student costs up to \$15,000 versus \$500 for a false positive. The model was trained on an 80/20 stratified split and serves as a screening tool requiring advisor review before any action is taken.

4.2 Stage 2: Prediction Model

Linear Regression was applied to the 115 dropout students to predict credits completed at withdrawal. Its interpretable coefficients allow advisors to distinguish early-dropout profiles from late-dropout profiles, directly supporting triage decisions. Results were validated against Excel Analysis ToolPak output.

5. Results and Key Findings

5.1 Stage 1: Classification Results

Metric	Result
Accuracy	73%
Precision (Dropout class)	47%
Recall (Dropout class)	30% — primary optimization target
F1-Score (Dropout class)	0.37
AUC-ROC	0.744

The Random Forest model (N=90 test set) achieved AUC-ROC of 0.744, confirming meaningful discriminatory ability. Recall of 30% represents a tangible improvement over no early-warning

system and can be increased by lowering the classification threshold. Current GPA (0.267) and Credit Completion Ratio (0.187) together account for over 45% of predictive power, with employment hours and attendance rate providing additional signal.

5.2 Stage 2: Prediction Results

Metric	Value
R ²	0.5753
Adjusted R ²	0.5517
Standard Error	9.55 credits
Significance F	4.32×10^{-18}

	Early Dropout (≤ 49 cr)	Late Dropout (> 49 cr)
Count	57 students	58 students
Avg GPA	2.03	2.47
Attendance Rate	73%	79%
Completion Ratio	0.48	0.61

Linear regression on the 115 dropout students explains 57.5% of variance in withdrawal timing ($R^2=0.575$). Two predictors are statistically significant: enrollment type (coef=+12.15, $p<0.001$) and employment hours (coef=+0.50, $p<0.001$). Splitting at the median (49 credits) reveals two distinct profiles — early dropouts (avg GPA 2.03, 73% attendance) require immediate academic intervention, while late dropouts (avg GPA 2.47, 79% attendance) are more likely leaving for financial or scheduling reasons.

5.3 Combined Interpretation

Stage 1 identifies who is at risk; Stage 2 determines urgency. Together they enable a tiered strategy: immediate high-touch outreach for early-dropout profiles, bi-weekly meetings for the broader high-risk group, and periodic monitoring for lower-urgency cases. All predictive findings are correlational — advisor judgment remains essential to every intervention decision.

6. Business Interpretation and Recommendations

The modeling results show that dropout risk is not evenly distributed and can be predicted using academic performance, engagement, and behavioral factors. To translate these findings into action, Caldwell University should adopt a targeted, data-driven intervention strategy.

Recommendation 1: Use the classification model to create a prioritized outreach list.

The university should use the classification model to identify and rank students by dropout risk, allowing Student Success teams to focus limited resources on the highest-risk students. This improves efficiency and is feasible because the model can be integrated into existing systems.

Recommendation 2: Use the prediction model to separate urgent from lower-urgency cases.

The prediction model should be used to estimate how far along students are likely to be at withdrawal, helping distinguish urgent cases from less urgent ones. This matters because early dropouts require faster intervention, and it is feasible since the model already provides these estimates.

Recommendation 3: Implement a tiered support model.

The university should adopt a tiered approach, including high-touch support for high-risk students, moderate support for mid-risk students, and monitoring for lower-risk students. This ensures resources are used efficiently and focuses effort where impact is greatest.

Recommendation 4: Embed the model into an operational workflow with clear ownership.

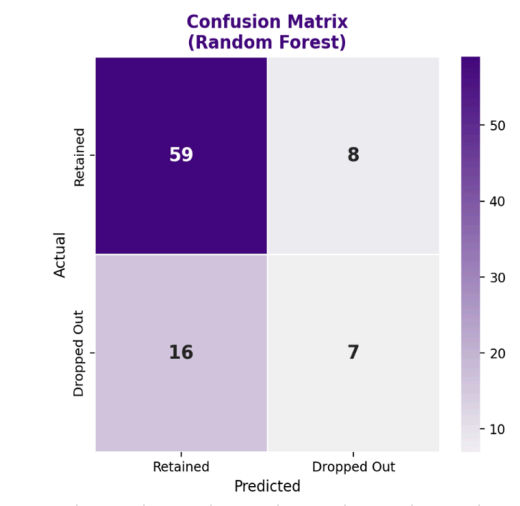
The process should be integrated across advising, financial aid, and mentoring teams, with clear ownership of follow-up actions. This ensures model insights are consistently acted on and is feasible using existing structures.

7. Expected Value Analysis

The goal of this analysis is to determine whether the targeting model creates business value for Caldwell University, not just predictive accuracy. Preventing a dropout is estimated to preserve \$15,000 in tuition, while intervention costs \$500 per student, based on benchmark estimates (College Board, 2025; Georgetown University Center on Education and the Workforce, 2025).

Because the model identifies risk but does not guarantee outcomes, a retention lift of 5%–20% is assumed. Based on the confusion matrix (7 true positives, 8 false positives, 16 false negatives, 59 true negatives), 15 students would receive intervention.

- **True positive (7):** Costs \$500, potential to save \$15,000
- **False positive (8):** Costs \$500, no return
- **False negative (16):** Missed student, loses \$15,000
- **True negative (59):** No cost, no impact



Expected value is calculated under three scenarios:

- **Conservative (5% lift):**

$$\text{Benefit} = 7 \times 5\% \times \$15,000 = \$5,250 \mid \text{Cost} = 15 \times \$500 = \$7,500 \mid \text{Net EV} = -\$2,250$$

- **Base (10% lift):**

$$\text{Benefit} = 7 \times 10\% \times \$15,000 = \$10,500 \mid \text{Cost} = \$7,500 \mid \text{Net EV} = \$3,000$$

- **Optimistic (20% lift):**

$$\text{Benefit} = 7 \times 20\% \times \$15,000 = \$21,000 \mid \text{Cost} = \$7,500 \mid \text{Net EV} = \$13,500$$

Expected value depends on the model threshold, balancing intervention cost and missed at-risk students. The model also misses 16 at-risk students, which may limit impact. The program is

worth piloting, as it generates positive value under realistic assumptions. However, if retention impact is low or model performance does not improve, it may not be worth scaling.

8. Measurement Plan

Caldwell should evaluate this program using both technical and business KPIs. Technical KPIs should include precision, recall, F1-score, and AUC-ROC to confirm that the classifier remains useful as a screening tool over time. Business KPIs should focus on outcomes tied to the Dean's decision, including dropout rate within the flagged cohort, retention lift relative to baseline, average credits preserved, intervention cost per retained student, and estimated tuition revenue protected. This distinction matters because a model can perform reasonably well on technical metrics without creating institutional value.

The baseline should be defined for the flagged high-risk cohort, not for the overall student population. Caldwell should document the historical dropout rate and average credits completed for students who would have been flagged as high risk before the new intervention begins. Model scores should refresh once per term, while intervention actions and outcomes should be reviewed monthly during the pilot period. The analytics team should maintain the scoring pipeline, Student Success leadership should review KPI performance, and advising, financial aid, and mentoring teams should document follow-up actions. A SMART outcome for the pilot is: *within one academic term, reduce dropout in the flagged high-risk cohort by 5 percentage points relative to baseline while maintaining positive expected value and documented follow-up for at least 90% of flagged cases.*

9. Pilot / Experimental Design

Improved outcomes after implementation would not, by themselves, prove that the program caused the improvement. If dropout declines after the model is introduced, Caldwell still would not know whether the change came from the intervention, normal semester variation, or differences in the students who received outreach. For that reason, the university should begin with a pilot before scaling the program more broadly.

The pilot should be conducted within the flagged high-risk cohort. Students in that cohort should be randomly assigned to either a treatment group, which receives the new tiered intervention strategy, or a control group, which continues under the university's normal support process. The primary KPI should be dropout rate within the flagged cohort by the end of the academic term. Secondary KPIs should include retention lift, average credits preserved, intervention completion rate, and net revenue protected after program cost.

Randomization matters because it reduces selection bias. Without it, staff may unintentionally prioritize students who already appear easier to retain, making the program look more effective than it really is. Caldwell should scale the program only if the treatment group shows a meaningful dropout reduction relative to control, positive expected value after intervention cost, and no major fairness concerns across student subgroups.

10. Risks, Limitations, and Ethical Considerations

This analysis has several limitations. The dataset does not include mental health, housing, family obligations, prior academic history, or advisor notes, so the model captures only part of the withdrawal process. The dropout class is smaller than the retained class, which makes recall harder to optimize, and the Stage 2 regression is based on only 115 dropout cases, which limits

precision. The results are also specific to Caldwell's current academic context and may not generalize beyond this setting.

The findings are predictive rather than causal. The model can identify patterns associated with dropout risk, but it cannot prove why students leave or which intervention will work best. There are also operational and ethical risks. If the threshold is set too high, Caldwell may miss vulnerable students; if it is set too low, staff capacity may be overwhelmed by false positives. First-generation status also requires careful handling because it may reflect structural disadvantage rather than individual-level risk. For that reason, the model should be used only as a human-in-the-loop screening and triage tool, with subgroup monitoring, documented threshold choices, and advisor oversight on every intervention decision.

11. Conclusion

Caldwell University faces rising student withdrawal rates and must decide how to allocate limited support resources to improve retention. The key decision is which students to prioritize for proactive intervention.

This report uses a two-stage approach: a classification model identifies at-risk students, and a prediction model estimates when they are likely to withdraw. Together, these models enable more targeted and timely interventions.

The recommended next step is to pilot the approach. Caldwell should use the model to guide outreach, track outcomes, and measure both operational and financial impact. The program should only be scaled if it demonstrates improved retention and positive expected value; otherwise, the model and strategy should be refined.

Resources

College Board. (2025). *Trends in college pricing and student aid 2025*.

https://research.collegeboard.org/media/pdf/Trends-in-College-Pricing-and-Student-Aid-2025-final_1.pdf

Georgetown University Center on Education and the Workforce. (2025). *Which programs most increase student success?*

<https://feed.georgetown.edu/access-affordability/which-programs-most-increase-student-success/>